

Multiple Choice (10 marks)

1. Statement 1| The original ResNets and Transformers are feedforward neural networks. Statement 2| The original Transformers use self-attention, but the original ResNet does not.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Which of the following points would Bayesians and frequentists disagree on?
 - (a) The use of a non-Gaussian noise model in probabilistic regression.
 - (b) The use of probabilistic modelling for regression.
 - (c) The use of prior distributions on the parameters in a probabilistic model.
 - (d) The use of class priors in Gaussian Discriminant Analysis.
5. Neural networks:
 - (a) Optimize a convex objective function
 - (b) Can only be trained with stochastic gradient descent
 - (c) Can use a mix of different activation functions
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: If the training loss increases with the number of epochs, it may indicate that the step size is too large.
7. True or False: The Stanford Sentiment Treebank contains movie reviews, which are used for sentiment analysis in natural language processing.
8. True or False: Principal Component Analysis (PCA) is not a supervised learning technique; it is an unsupervised method used for dimensionality reduction.
9. True or False: Using strong classifiers in bagging effectively reduces the risk of overfitting during model training.
10. True or False: Linear hard-margin SVM can only be used when the training data is linearly separable.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to replace maximum n occurrences of spaces, commas, or dots with a colon.
12. Write a function to perform the exponentiation of the given two tuples.

End of Paper